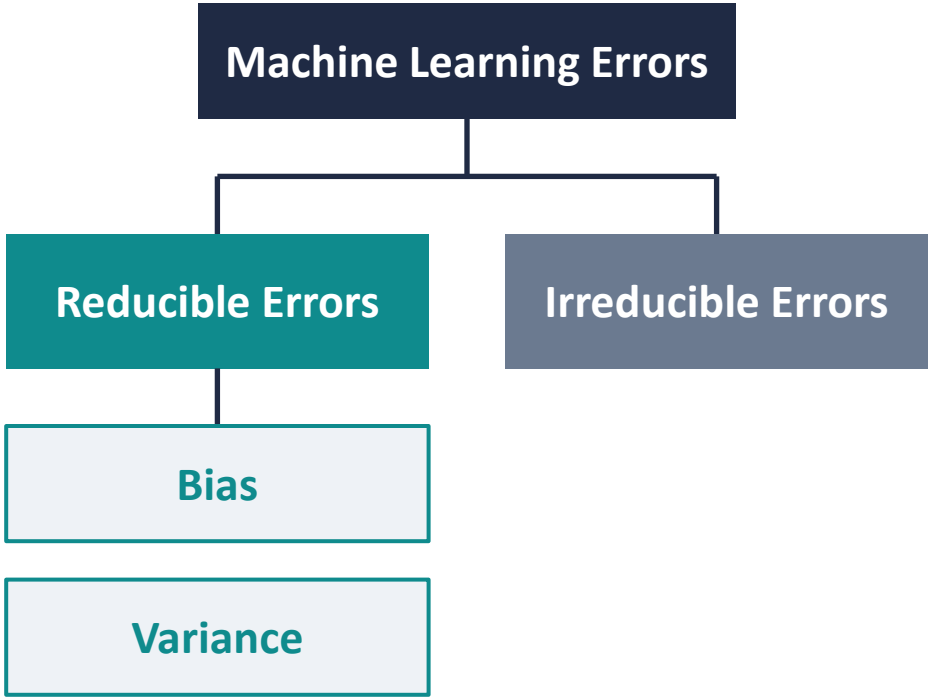
Bias and Variance

Machine Learning (ML): reducible errors explained

Machine Learning Errors



Bias and Variance

- Reducible errors
- Helps in parameter tuning
- Helps in deciding a better fitted model

Bias

Bias is the difference between the actual value and the predicted value.

Low Bias

- Few assumptions
- Model closely matches the training dataset

High Bias

- More assumptions
- Model does not closely match the training dataset
- High error rate

Variance

Variance is the spread of data from its mean position.

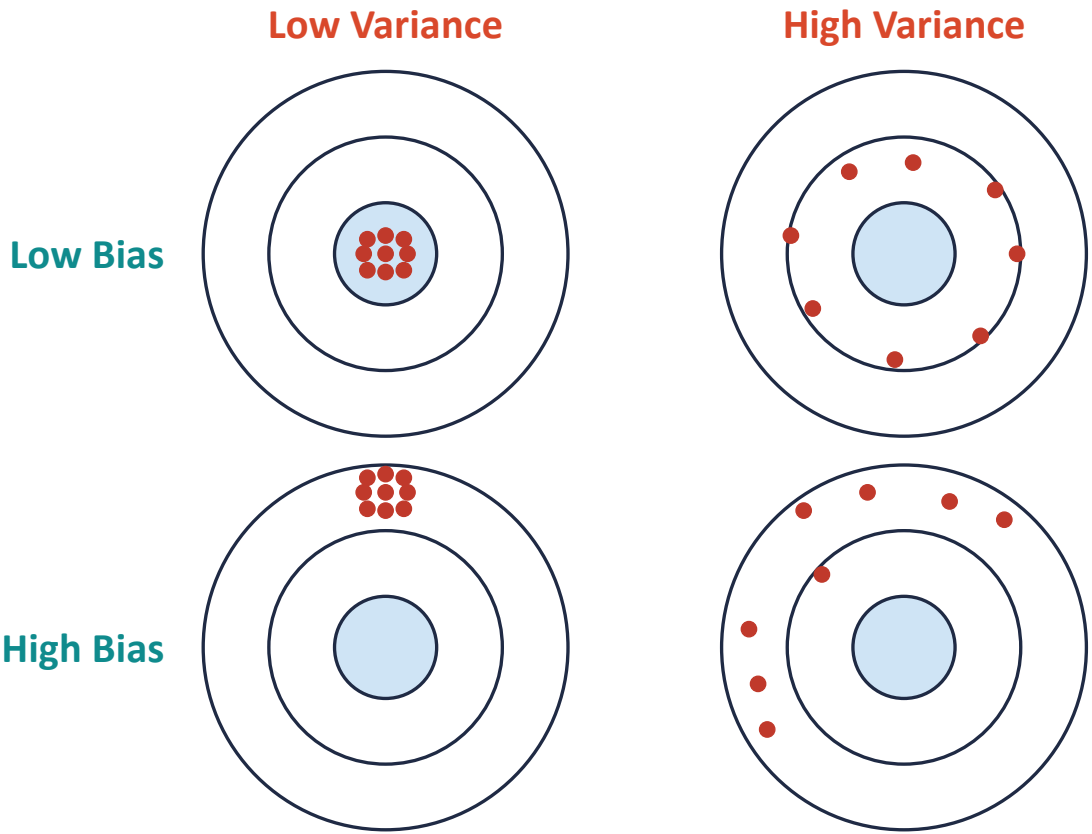
Low Variance

- Model is less sensitive to changes in training data
- Produces consistent estimates on different data

High Variance

- Model is very sensitive to changes in training data
- Produces significant changes in estimates on different data

Bias and variance: the dartboards



Total prediction error

Total Error = Bias + Variance + Irreducible Noise

Bias

Too simple, wrong assumptions. Underfits.

Variance

Too sensitive to training data. Overfits.

Noise

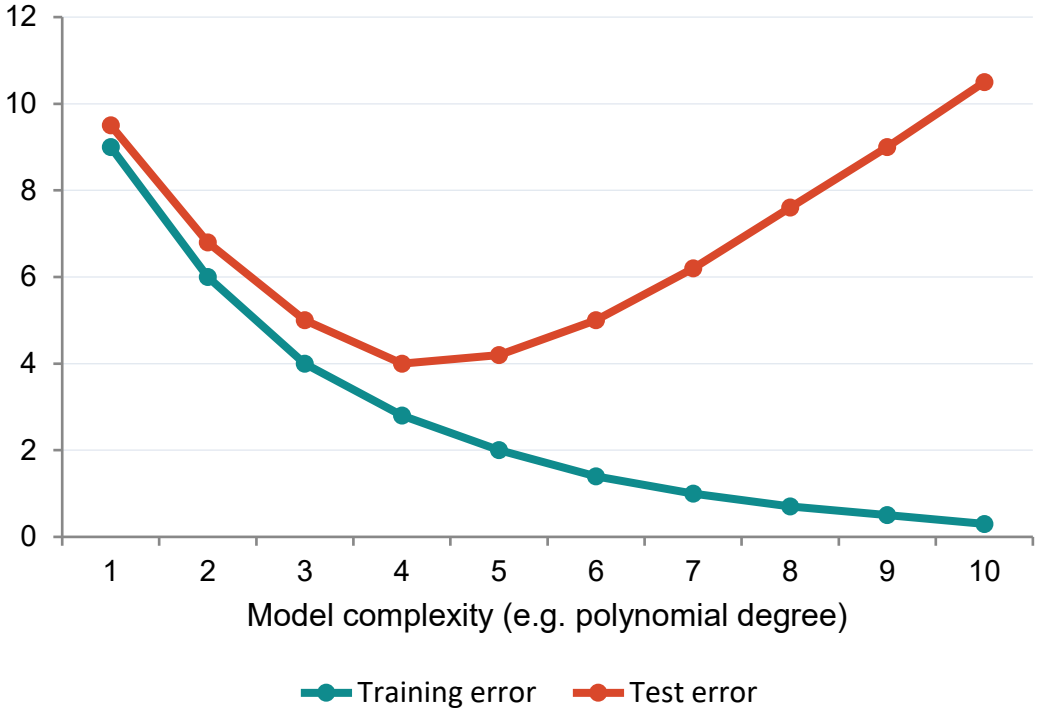
Randomness no model can remove.

Example: three students preparing for an exam

Student	Strategy	Result
High bias	Studies only chapter headings	Poor on practice and real exam (underfitting)
High variance	Memorizes every practice question	100% on practice, fails when wording changes (overfitting)
Balanced	Understands concepts, practices varied questions	Does well on both

Other examples: a straight line fitted to curved data (high bias); a very deep decision tree (high variance).

Complexity vs. error: the U-shaped curve

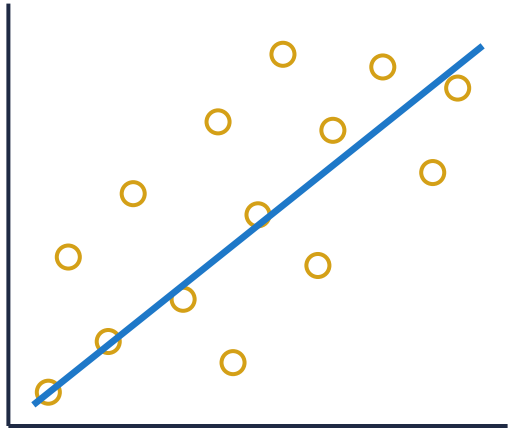


Left side
High bias: both errors high (underfit)

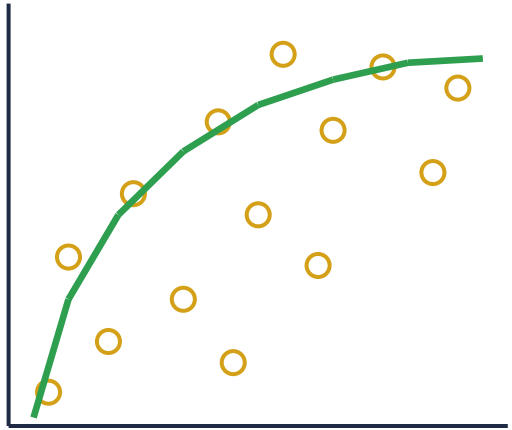
Bottom of the U
Best model: lowest test error

Right side
High variance: train low, test high (overfit)

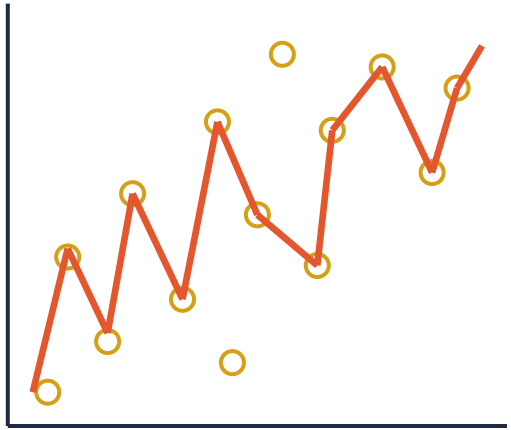
Underfitting, best fit and overfitting



Underfitting



Best fitting



Overfitting

Y = output, X = input. Circles are the data points; the line is the model's fit.

Underfitting vs. overfitting

	Underfitting	Overfitting
Model	Not complex	Complex
Error type	High bias	High variance
Training set	Not accurate	Accurate
Features	Less number of features	Large number of features
Training	Less training	Large training

Analogy: is it a ball?

Features the model may learn: Sphere, Eat, Play, Size

Underfitting

Knows too few features, so an orange (also a sphere) is wrongly called a Ball.

Best fitting

Learns the essential idea of a ball, so it labels new objects correctly.

Overfitting

Learns too many exact features, so a cricket ball that differs slightly is wrongly called Not Ball.

Diagnose and fix

Problem	Signs	Fixes
High bias (underfitting)	Train and test error both high	More complex model, add features, less regularization, boosting
High variance (overfitting)	Train error low, test error high	More data, simpler model, regularization, pruning, bagging

Ensembles: bagging (Random Forest) mainly reduces variance; boosting mainly reduces bias.

Class activity: bias or variance?

1. Train accuracy 60%, test accuracy 58%
2. Train accuracy 100%, test accuracy 65%
3. Decision tree with max_depth = 1
4. 1-nearest-neighbor classifier

Key takeaways

- Bias = too simple (underfit). Variance = too sensitive (overfit).
- Bias and variance are reducible errors; noise is not.
- More complexity lowers bias but raises variance.
- Judge a model by test error, never training error alone.
- Balance them with regularization, more data, and ensembles.